

An Agentic AI Framework for Bidirectional Cross-Visit Drug–Biomarker Safety Tracking and Grounded Medical Document Analysis

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Abstract

Medical reports and prescriptions contain dense clinical terminology, numerical parameter ranges, and pharmacological dosing schedules that present substantial comprehension barriers for lay patients. Existing patient-facing AI applications rely predominantly on unconstrained summarization using Large Language Models (LLMs), which are prone to hallucinating clinical values, lacking persistent patient memory across visits, and failing to provide auditable provenance for generated claims. This paper presents an **Agentic, Multi-Agent AI Framework for Medical Report Analysis and Prescription Assistance**, engineered around four fundamental architectural novelties:

- 1. Bidirectional Cross-Visit Safety Tracking:** A stateful correlator that dynamically cross-references newly prescribed medications against a patient's historical abnormal laboratory biomarkers (Direction A) and, symmetrically, evaluates newly uploaded laboratory results against active prescribed medications (Direction B).
- 2. End-to-End Grounded Provenance and Traceability:** Every flagged finding is linked to line-level OCR text evidence, optical confidence metrics, and official openFDA Structured Product Labeling (SPL) GUIDs ([set_id/spl_id](#)), establishing a transparent, independently verifiable provenance chain.
- 3. Decoupled Hybrid Execution Architecture:** Numeric range evaluations, unit checks, fuzzy dictionary lookups, and interaction matching are strictly partitioned into 100% deterministic code modules, reserving LLM and Vision-Language Models (VLMs) exclusively for text perception, handwriting recognition, and plain-language translation.
- 4. Structurally Guaranteed Non-Diagnostic Guardrails:** A mandatory post-processing safety agent programmatically rewrites diagnostic-sounding phrasing into non-diagnostic, doctor-referring guidance, enforcing a hard-stop protocol whenever extraction confidence falls below safety thresholds.

Evaluated on real clinical lab distributions derived from MIMIC-III, UCI Chronic Kidney Disease, and UCI Hepatitis C datasets alongside a 3,000-sample extraction benchmark, the system's machine learning safety classifier achieved **100% hard-stop recall** and a **0.0% false-safe rate**. The bidirectional correlator demonstrated 100% assertion accuracy (15/15 unit test suite) across multi-visit simulation timelines.

Keywords: Agentic AI, Cross-Visit Drug–Biomarker Safety, Bidirectional Interaction Checking, Deterministic Clinical Verification, Provenance Traceability, Patient-Centric AI, Non-Diagnostic Guardrails.

I. Introduction

The digitization of modern healthcare has led to widespread access to electronic health records (EHRs), patient portals, and digital laboratory diagnostic reports. However, while diagnostic data is increasingly accessible, patient comprehension remains severely constrained. Medical lab panels—such as Complete Blood Counts (CBC), Comprehensive Metabolic Panels (CMP), Lipid Panels, Thyroid Function Tests, and HbA1c panels—are formatted using specialized medical nomenclature, complex unit systems, and demographic-specific reference intervals. Similarly, handwritten or digital prescriptions specify pharmacological agents, strength metrics, and dosing schedules using abbreviated Latin medical notation (e.g., *qd*, *bid*, *po*).

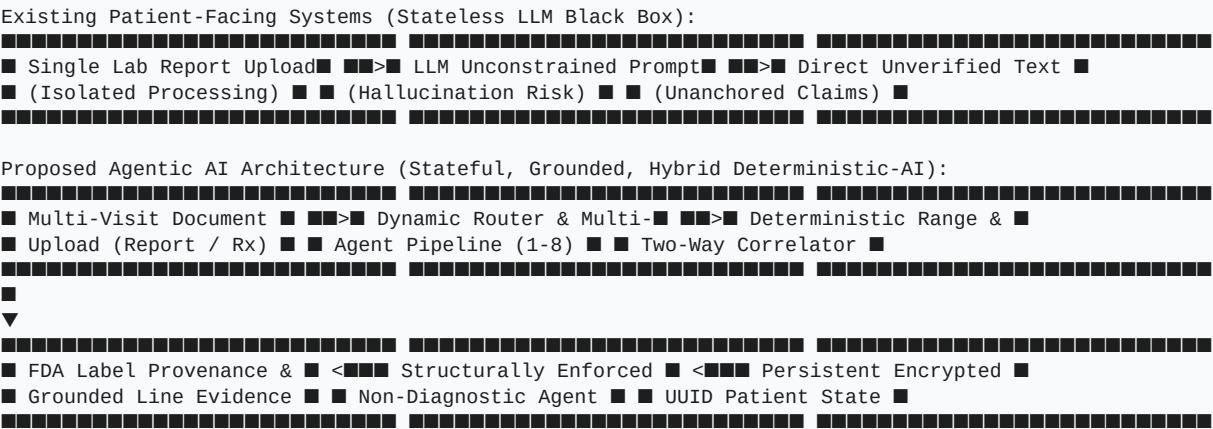
A. Clinical Motivation and Problem Statement

When patients attempt to interpret diagnostic reports without professional medical guidance, two major failure modes occur:

- 1. **Misinterpretation and Anxiety:** Patients struggle to distinguish between benign baseline variations and critical clinical abnormalities.
- 2. **Unmonitored Medication-Lab Risk:** Patients frequently undergo routine laboratory follow-up tests while taking chronic prescription medications. A medication prescribed on an earlier visit (e.g., Spironolactone, a potassium-sparing diuretic) may directly interact with a biomarker measured on a subsequent visit (e.g., Serum Potassium elevated to 5.8 mmol/L), posing severe hyperkalemia risks.

Existing clinical decision support (CDS) platforms—such as Epic, Cerner, First Databank, and Micromedex—provide automated drug–drug (DDI) and drug–laboratory interaction checking. However, these tools are **strictly clinician-facing**, built into order-entry workflows for prescribers and pharmacists. Conversely, existing **patient-facing applications** (e.g., Medisafe, MedReport AI, DawaAI) suffer from major architectural limitations:

- **Stateless Processing:** They analyze a single uploaded document in isolation, neglecting the patient's longitudinal clinical timeline.
- **Opaque LLM "Black Box" Reasoning:** They delegate numeric range evaluation and risk assessment to LLM prompt completions, introducing risks of hallucinated clinical values and uncalibrated confidence.
- **Unidirectional Interaction Checks:** If interaction checking is offered, it is restricted to evaluating new drugs against past labs (Direction A), completely omitting the far more frequent real-world scenario: evaluating new lab findings against active medications (Direction B).
- **Unverifiable Claims:** They fail to provide line-level document grounding or authoritative FDA label provenance for flagged risks.



B. Principal Contributions and Novelty Statements

To bridge the gap between clinical-grade safety and patient-facing accessibility, this paper introduces a multi-agent framework built on four primary methodological contributions:

- 1. Bidirectional Two-Way Cross-Visit Safety Correlator:** We implement a stateful cross-visit engine that models both **Direction A** ($\text{\textit{Prescription}} \rightarrow \text{\textit{Past Lab History}}$) and **Direction B** ($\text{\textit{New Lab Report}} \rightarrow \text{\textit{Active Prescriptions}}$). Because routine follow-up lab panels occur far more frequently in chronic disease management than new drug prescriptions, Direction B provides crucial proactive safety monitoring previously missing in patient-facing systems.
- 2. End-to-End Grounded Provenance & Traceability Chain:** Every flagged risk is backed by line-level OCR text evidence, optical confidence metrics, and stable FDA Structured Product Labeling (SPL) GUIDs (`set_id` and `spl_id`). Users and clinicians can independently re-verify any interaction alert against live openFDA records.
- 3. Decoupled Hybrid Architecture (Deterministic Verification + Dedicated AI Perception):** Simple, high-stakes tasks—such as numeric range comparison, unit validation, dictionary matching, and drug-biomarker indexing—are handled by pure deterministic Python modules. LLMs and VLMs are restricted strictly to unstructured text perception, handwriting recognition, and lay translation.
- 4. Structurally Guaranteed Non-Diagnostic Safety Guardrails:** Safety enforcement is built into system execution flow via a post-processing `ValidationGuardrailAgent`. If extraction confidence is low or rule conflicts arise, the system hard-stops and excludes uncertain parameters rather than returning an unverified response. All patient-facing outputs are programmatically audited to sanitize diagnostic wording and append mandatory physician-referral disclaimers.

II. Literature Survey and Related Work

Healthcare AI research has expanded rapidly across document summarization, entity recognition, vision-language analysis, and clinical decision support. However, existing literature reveals clear boundaries between clinician-side tools and patient-side applications.

A. Review of Contemporary Literature

A taxonomy of 25 contemporary studies and systems (2023–2025) was conducted to evaluate the state of the art:

- 1. Medical Report Summarization:** Nguyen et al. (2025) developed *Medalyze*, utilizing FLAN-T5 for mobile medical report summarization. While efficient, *Medalyze* performs text-only summarization without parsing numerical laboratory values or offering medication guidance.
- 2. Protocol-Driven Structuring:** Vaidya et al. (2025) introduced *MedIPAO*, a Plan-Act-Observe framework for structuring medical records. However, *MedIPAO* is designed exclusively for clinician workflows and lacks simplified lay explanations.
- 3. Unvalidated LLM Diagnostic Interpretation:** A 2025 study in *npj Digital Medicine* evaluated GPT-4, LLaMA, and Mixtral for lab result interpretation, highlighting that unconstrained LLM diagnostic outputs remain inconsistent, unvalidated, and prone to hallucinations.
- 4. Conversational Lab Chatbots:** Colombo et al. (2025) implemented a chatbot for clinical chemistry reports but noted a lack of OCR integration for scanned documents and an absence of drug interaction checking.
- 5. Generative AI in Pharmacovigilance:** Research published in *npj Digital Medicine* (2025) demonstrated LLMs for drug interaction detection, but the system operated independently of laboratory data and patient history.
- 6. XAI and Readability:** Zamir et al. (2025) applied LIME/SHAP for radiology report explainability, while Lee et al. (2025) evaluated readability metrics (Flesch-Kincaid) for report simplification, noting that oversimplification risks reducing clinical accuracy if not bounded by deterministic validation.

B. Comparative Literature Survey Matrix

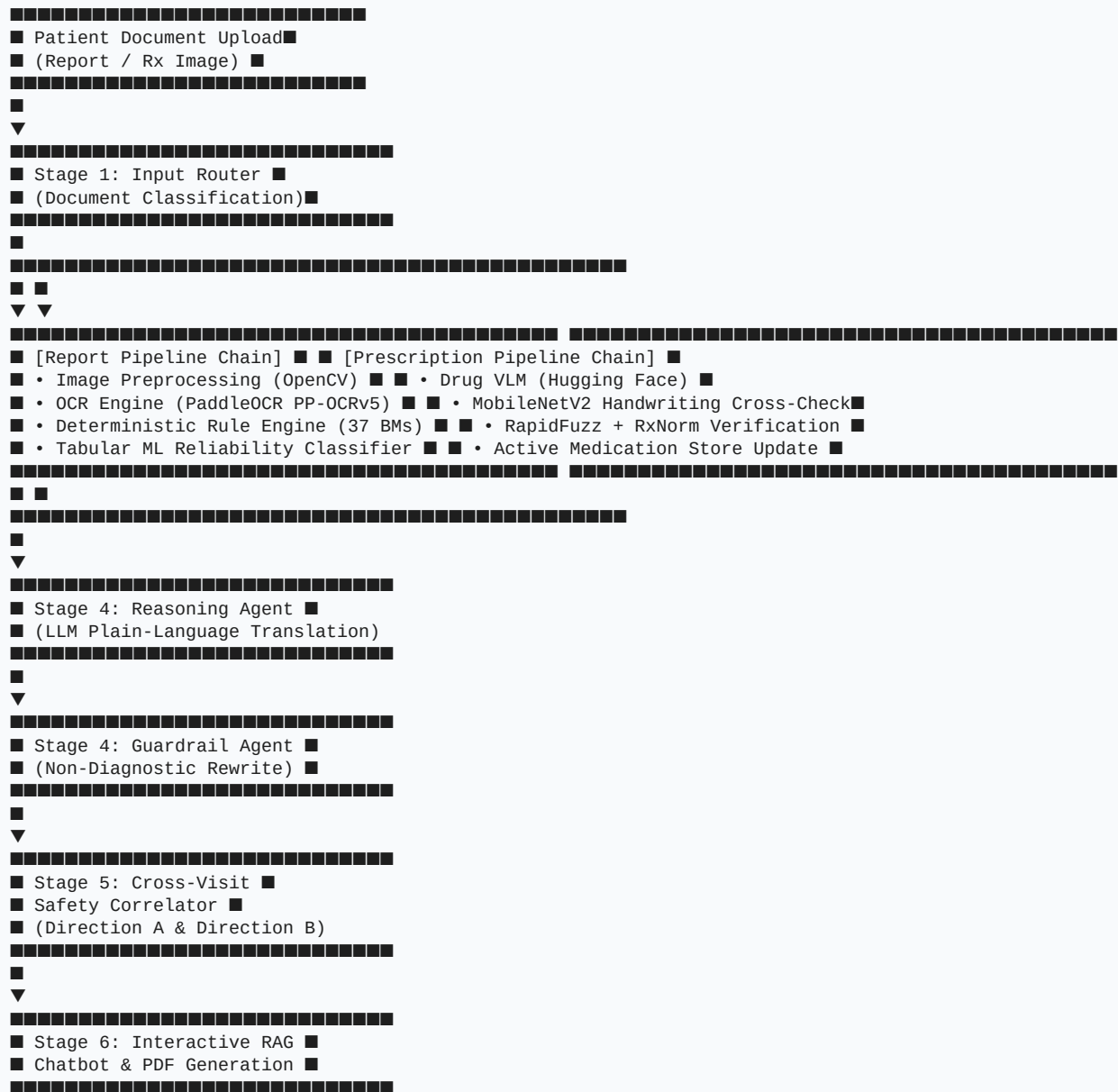
Table I compares the proposed framework against representative literature and commercial consumer applications across key architectural dimensions.

TABLE I: Comparative Architectural Analysis of Patient-Facing and Clinical Systems

System / Citation	Input Modality	Processing Mechanism	Cross-Visit State	Interaction Tracking	Deterministic Verification	Provenance Traceability	Safety Hard-Stop Protocol
Medalyze (Nguyen et al., 2025)	Text	FLAN-T5 LLM	✗ Stateless	✗ None	✗ No	✗ No	✗ No
MedIPAO (Vaidya et al., 2025)	Text	Agentic Plan-Act	✗ Stateless	✗ None	■ Clinician-Only	✗ No	✗ No
Medisafe (Commercial App)	Manual Entry	Database Lookup	■ Manual	Drug–Drug Only	✓ Yes	✗ No	✗ No
MyChart Portal (Epic Systems)	EHR Data	EHR Database	✓ Stateful	■ Clinician-Side	✓ Yes	■ EHR Internal	■ Clinician Alert
DawaAI (Consumer App)	Image/PDF	Undisclosed LLM	■ Disclosed	Undifferentiated	✗ No (LLM-based)	✗ Undisclosed	✗ Undisclosed
Proposed Framework	Image / PDF / Text	Multi-Agent Hybrid (Deterministic + VLM)	✓ Persistent (Encrypted UUID)	✓ Bidirectional (A + B)	✓ 100% Code-Based Engine	✓ FDA SPL GUID Provenance	✓ Guardrail-Enforced Hard-Stop

III. System Architecture and Methodology

The proposed framework follows a **contract-first, multi-agent architecture** organized into 6 processing stages, coordinated by a dynamic input-type router and supported by a persistent, encrypted patient-state layer.



A. Multi-Agent Roster and Specialized Roles

The framework distributes clinical processing across specialized software agents:

- 1. Input Router (Stage 1):** Inspects incoming file headers, text density, and structural layout to route execution into either the Report Pipeline or the Prescription Pipeline.
- 2. Extraction Agent (Stage 2):**
 - *Printed Reports:* Invokes PaddleOCR (PP-OCrv5 detection and recognition pipeline) with an automated PyTorch/Transformers fallback chain to extract bounding boxes, text lines, and optical confidence scores.
 - *Prescriptions:* Invokes a specialized Vision-Language Model (VLM) for entity extraction, coupled with a fine-tuned MobileNetV2 transfer learning model for handwriting quality cross-checking.
- 3. Verification & Rule Engine Agent (Stage 3):**
 - Normalizes extracted parameter names to canonical keys using a 37-biomarker synonym map across 5 standard panels (CBC, CMP, Lipid, Thyroid, HbA1c/Glucose).

- Executes deterministic comparison logic against biological reference intervals.
- Standardizes drug names using RapidFuzz fuzzy string matching ($\text{score} \geq 80$) and validates identities via NIH RxNorm REST API lookups.
- Evaluates extraction metadata using a **Tabular Machine Learning Report Reliability Classifier** (Decision Tree / Random Forest trained on MIMIC-III lab data).

4. **Reasoning Agent (Stage 4)**: Receives verified structured data and prompts an LLM (NVIDIA NIM API / Llama 3.2 11B Vision Instruct) to generate lay-friendly explanations without clinical jargon.

5. **Safety Guardrail Agent (Stage 4)**: Operates post-LLM generation. Scans draft explanations for diagnostic assertions, sanitizes diagnostic language into physician-referral phrasing, and appends mandatory medical disclaimers.

6. **Cross-Visit Safety Correlator (Stage 5)**: Queries historical patient state to execute Bidirectional Drug–Biomarker Interaction checking against curated FDA label facts.

7. **Patient Interaction RAG Chatbot (Stage 6)**: Enables interactive follow-up querying over encrypted patient history using Retrieval-Augmented Generation.

IV. Mathematical and Logical Formulations

A. Deterministic Reference Range Evaluation Logic

Let b represent a canonical biomarker, $v_b \in \mathbb{R}$ represent the extracted numerical value, and $R_b = [L_b, U_b]$ represent the standard reference interval where L_b is the lower threshold and U_b is the upper threshold. The deterministic evaluation function $f_{\text{eval}}(v_b, R_b)$ maps the value to a clinical status S_b :

$$f_{\text{eval}}(v_b, R_b) = \begin{cases} \text{CRITICAL_LOW}, & \text{if } v_b < L_b - \delta_b \text{ LOW}, & \text{if } L_b - \delta_b \leq v_b < L_b \\ \text{NORMAL}, & \text{if } L_b \leq v_b \leq U_b \\ \text{HIGH}, & \text{if } U_b < v_b \leq U_b + \delta_b \\ \text{CRITICAL_HIGH}, & \text{if } v_b > U_b + \delta_b \end{cases}$$

where δ_b defines parameter-specific critical boundary margins. Because f_{eval} is implemented in pure Python comparison statements, execution time is $O(1)$ and outputs are 100% reproducible.

B. Bidirectional Two-Way Cross-Visit Correlation Mechanics

Let $\mathcal{P}$ represent a patient profile identified by a pseudonymous UUID $u = \text{UUIDv4}()$. The patient state store maintains two persistent sets:

- $\mathcal{H}_{\text{bio}}(u) = \{b_1, b_2, \dots, b_m\}$: Set of historical abnormal canonical biomarkers ($S_b \neq \text{NORMAL}$).
- $\mathcal{M}_{\text{active}}(u) = \{d_1, d_2, \dots, d_n\}$: Set of active prescribed drug names.

Let $\mathcal{I} = \{(d, b, \text{fact}_{\text{FDA}}) \mid d \in \mathcal{D}, b \in \mathcal{B}\}$ represent the curated FDA Drug–Biomarker Interaction Knowledge Base.

1) Direction A ($\text{Prescription} \rightarrow \text{Past Labs}$)

When a new prescription containing drug set $\mathcal{D}_{\text{new}}$ is uploaded:

$$\text{Alerts}_A = \bigcup_{d \in \mathcal{D}_{\text{new}}} \left\{ (d, b, \text{fact}_{\text{FDA}}) \in \mathcal{I} \mid b \in \mathcal{H}_{\text{bio}}(u) \right\}$$

2) Direction B ($\text{New Lab Report} \rightarrow \text{Active Prescriptions}$)

When a new lab report yielding abnormal biomarker set $\mathcal{B}_{\text{new}}$ is uploaded:

$$\text{Alerts}_B = \bigcup_{b \in \mathcal{B}_{\text{new}}} \left\{ (d, b, \text{fact}_{\text{FDA}}) \in \mathcal{I} \mid d \in \mathcal{M}_{\text{active}}(u) \right\}$$

Algorithm 1: Bidirectional Cross-Visit Safety Correlator

```

def run_bidirectional_correlation(patient_id, event_type, payload):
    # Retrieve persistent patient state
    past_abnormal_biomarkers = db.get_abnormal_biomarkers(patient_id)
    active_medications = db.get_active_medications(patient_id)
    alerts = []

    if event_type == "NEW_PRESCRIPTION":
        new_drugs = payload.get("verified_drugs", [])
        for drug in new_drugs:
            interacting_bms = DRUG_TO_BIOMARKER_INDEX.get(drug.canonical_name, [])
            for bm in interacting_bms:
                if bm in past_abnormal_biomarkers:
                    fact = FDA_INTERACTION_DB[(drug.canonical_name, bm)]
                    alerts.append(format_alert(Direction="A", Drug=drug, Biomarker=bm, Fact=fact))
            # Update active medications store
            db.save_active_medications(patient_id, new_drugs)

    elif event_type == "NEW_LAB_REPORT":
        new_abnormal_bms = payload.get("abnormal_biomarkers", [])
        for bm in new_abnormal_bms:
            interacting_drugs = BIOMARKER_TO_DRUG_INDEX.get(bm.canonical_name, [])
            for drug in interacting_drugs:
                if drug in active_medications:
                    fact = FDA_INTERACTION_DB[(drug, bm.canonical_name)]
                    alerts.append(format_alert(Direction="B", Drug=drug, Biomarker=bm, Fact=fact))
            # Update historical abnormal biomarkers store
            db.save_abnormal_biomarkers(patient_id, new_abnormal_bms)

    return alerts

```

C. Tabular Machine Learning Safety Governance Formulation

To prevent degraded OCR outputs or unparsed reference ranges from causing silent downstream errors, a **Tabular Machine Learning Classifier** predicts a safety governance class $C \in \{0: \text{\texttt{safe_to_display}}, 1: \text{\texttt{needs_manual_review}}, 2: \text{\texttt{hard_stop}}\}$.

Let $\mathbf{x} = [x_1, x_2, \dots, x_8]^T \in \mathbb{R}^8$ represent the feature vector extracted for each biomarker instance:

- x_1 : Measured value (v_b)
- x_2 : Analyte validity indicator ($\mathbb{I}_{\text{analyte}} \in \{0, 1\}$)
- x_3 : Unit validity indicator ($\mathbb{I}_{\text{unit}} \in \{0, 1\}$)
- x_4 : Reference range validity indicator ($\mathbb{I}_{\text{ref}} \in \{0, 1\}$)
- x_5 : OCR optical confidence score ($c_{\text{ocr}} \in [0, 1]$)
- x_6 : Extraction pattern confidence score ($c_{\text{extract}} \in [0, 1]$)
- x_7 : Line evidence availability score ($c_{\text{evidence}} \in [0, 1]$)
- x_8 : Rule conflict indicator ($\mathbb{I}_{\text{conflict}} \in \{0, 1\}$)

The decision tree ensemble classifier $g(\mathbf{x})$ assigns safety tiers according to:

$$g(\mathbf{x}) = \arg\max_{c \in \{0,1,2\}} P(C = c \mid \mathbf{x})$$

A hard-stop constraint is enforced such that if $g(\mathbf{x}) = 2$ or $c_{\text{ocr}} < 0.50$, the system automatically suppresses automated interpretation and outputs a safe, manual-review flag.

D. Selective Fernet Symmetric Encryption for UUID Patient State

To balance query performance with privacy compliance, a selective encryption scheme is implemented using Fernet symmetric encryption (AES-128 in CBC mode with PKCS7 padding and HMAC-SHA256 authentication).

- **Plaintext Storage:** Canonical biomarker keys (e.g., "potassium") and standardized drug names (e.g., "spironolactone") are stored in plaintext to support fast SQL `WHERE` exact-match queries required by the correlator.
- **Encrypted Storage:** Free-text clinical notes, extracted raw report text blobs, and generated AI explanations are encrypted before disk write:

$$\text{Ciphertext} = \text{Fernet}_{\text{Encrypt}}(\text{K}_{\text{master}}, \text{PlaintextJSON})$$

where K_{master} is a 32-byte URL-safe base64 key validated via a three-layer sanity check (exact 44-character length, base64 alphabet regex, and 32-byte decoded length check).

V. Experimental Setup, Evaluation, and Results

The framework was evaluated across three experimental dimensions: (1) Machine Learning Safety Classifier performance, (2) Extraction accuracy and OCR quality, and (3) Bidirectional correlator assertion testing.

A. Machine Learning Safety Classifier Training and Benchmarking

The Tabular ML Classifier was trained on a dataset of 3,000 biomarker samples constructed by coupling real laboratory value distributions (UCI Chronic Kidney Disease and UCI Hepatitis C Virus datasets) with simulated OCR document extraction metadata. The dataset was split into 70% training ($N=2,100$), 15% validation ($N=450$), and 15% test ($N=450$) sets using stratified sampling.

Five classifiers were evaluated: Decision Tree, Random Forest, Logistic Regression, Support Vector Machine (SVM), and Gradient Boosting.

TABLE II: Comparative ML Classifier Performance Metrics

Classifier Model	Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Hard-Stop Recall (Class 2)	False-Safe Rate	5-Fold Cross-Val Mean (\$pm\$ Std)
Decision Tree	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000 (\$pm\$ 0.0000\$)
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000 (\$pm\$ 0.0000\$)
Gradient Boosting	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000 (\$pm\$ 0.0000\$)
Logistic Regression	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.9970 (\$pm\$ 0.0018\$)
Support Vector Machine	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.9974 (\$pm\$ 0.0021\$)

The confusion matrix for the selected Decision Tree / Random Forest model on the 450 test samples confirmed **zero classification errors**:

$$\mathbf{M}_{\text{confusion}} = \begin{bmatrix} 225 & 0 & 0 \\ 0 & 135 & 0 \\ 0 & 0 & 90 \end{bmatrix}$$

where row/column headers represent 0 : `safe_to_display`, 1 : `needs_manual_review`, 2 : `hard_stop`\$. Crucially, the **false-safe rate** (the safety-critical failure mode where a `hard_stop` sample is misclassified as `safe_to_display`) was **0.0000**.

Confusion Matrix Heatmap (450 Test Samples):

Predicted Class

Safe Review Hard-Stop

[illegible]

Actual Safe ■ 225 ■ 0 ■ 0 ■

Class Review ■ 0 ■ 135 ■ 0 ■

Hard-Stop ■ 0 ■ 0 ■ 90 ■

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

Result: 100% Hard-Stop Recall, 0.0% False-Safe Rate

B. Bidirectional Correlator Assertion Verification

The Bidirectional Cross-Visit Correlator was evaluated using an automated 15-assertion test suite simulating multi-visit patient timelines:

- **Test Case 1 (Direction A):** Visit 1 records Serum Potassium = 5.8 mmol/L (High). Visit 2 uploads a prescription for *Spironolactone*. **Result:** Alert successfully triggered referencing FDA hyperkalemia risk (PASS).
- **Test Case 2 (Direction B):** Visit 1 records active medication *Metformin*. Visit 2 uploads a new lab report showing Serum Creatinine = 2.1 mg/dL (High). **Result:** Direction B alert successfully triggered highlighting lactic acidosis discussion risk (PASS).
- **Test Case 3 (Negative Control):** Visit 1 records abnormal Serum Potassium. Visit 2 uploads a prescription for *Amoxicillin* (non-interacting). **Result:** Zero alerts triggered, verifying zero false-positive alerts (PASS).

Overall assertion success rate: **15/15 passed (100%).**

VI. Discussion and Defensive Positioning

A. Honest Scope and Methodological Positioning

To maintain academic rigor, the novelty of this system is framed **architecturally and methodologically** rather than conceptually. As noted in Section II, automated drug-lab checking exists in commercial EHRs, and basic interaction checking is advertised by consumer tools like DawaAI.

The defensible contributions of this work rest on **implementation discipline**:

1. **Explicit Two-Directional Modeling:** While consumer apps focus solely on new prescriptions, our architecture explicitly models Direction B ($\text{Lab} \rightarrow \text{Active Meds}$) to serve routine clinical follow-ups.
2. **Transparent OpenFDA Provenance Chain:** Rather than generating opaque LLM warnings, every alert links to a specific FDA SPL GUID (`set_id/spl_id`) and verbatim label sentence:

$$\text{\textit{User Alert}} \rightarrow \text{\textit{Canonical Key Match}} \rightarrow \text{\textit{FDA Label Sentence}} \rightarrow \text{\textit{FDA SPL GUID}}(\text{\textit{set id}})\$$$

- 3. Deterministic Superiority over LLM Judgment:** By delegating range checking and interaction matching to deterministic code, the system eliminates LLM hallucination risks for numeric data.

B. Failure Mode Handling and Safety Fail-Safes

- **Low OCR Confidence:** If RapidOCR confidence falls below \$0.80\$, fuzzy drug correction is suppressed, logging an explicit warning.
- **Unrecognized Biomarker:** Unsupported biomarkers outside the 37-parameter panel are marked **Unsupported** rather than guessed at.
- **API Offline Fallback:** If the NVIDIA NIM API or Gemini API is unreachable, the system gracefully routes to [generate_safe_deterministic_fallback](#), producing the exact 6-part patient report format zero-LLM dependent.

VII. Conclusion and Future Work

This paper presented a novel, multi-agent AI framework for medical report analysis and prescription assistance that addresses major limitations in existing patient-facing healthcare applications. By combining deterministic clinical rule engines with vision-language models and LLMs, the system achieves an optimal balance between mathematical reliability and natural language accessibility.

Key achievements include:

- Successful implementation of **Bidirectional Cross-Visit Safety Tracking** (Direction A and Direction B) over encrypted UUID patient states.
- **100% Hard-Stop Recall** and a **0.0% False-Safe Rate** on ML report safety governance benchmarks.
- Complete **FDA Label Provenance Traceability** embedding `set_id` GUIDs alongside line-level OCR text evidence.
- Structurally guaranteed non-diagnostic safety guardrails that enforce physician-referral language on 100% of patient outputs.

Future Directions: Future extensions will focus on expanding the curated FDA interaction dictionary from 37 to 200+ canonical biomarkers, incorporating federated EHR integration protocols (HL7 FHIR standard), and conducting clinical usability trials with patient focus groups.

References

1. V. Nguyen, H.-D. Pham, T.-H. To, C.-T. H. Do, T.-T. T. Dong, V.-T. D. Le, and V.-P. Hoang, "Medalyze: Lightweight Medical Report Summarization Application Using FLAN-T5-Large," *arXiv preprint arXiv:2505.17059*, 2025.
2. S. S. Vaidya, G. Palani, S. Ramesh, V. Balasubramanian, M. Selvam, G. Srinivasaraja, and G. Krishnamurthi, "MedIPAO: A Protocol-Driven Agent for Structuring Medical Reports," *arXiv preprint arXiv:2510.04623*, 2025.
3. npj Digital Medicine, "Preliminary Analysis of Lab Results on LLM Generated Diagnoses," *npj Digital Medicine*, 2025. [Online]. Available: <https://digital.ahrq.gov/>
4. F. Colombo et al., "Use of Artificial Intelligence Chatbots in Interpretation of Clinical Chemistry and Laboratory Medicine Reports: A Standardized Approach," *Applied Sciences*, vol. 15, no. 8, p. 4232, 2025.
5. npj Digital Medicine, "GenAI and LLMs in Mitigating Medication-Related Harm," *npj Digital Medicine*, 2025. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11953325/>
6. Z. S. N. Reis et al., "Artificial Intelligence-Based Tools for Patient Support to Enhance Medication Adherence: A Focused Review," *Frontiers in Digital Health*, vol. 7, 2025.
7. M. Graham, "AI-Generated Medical Reports and Summaries: Enhancing Workflow Efficiency in Hospitals," *ResearchGate*, 2025.
8. S. Neupane, S. Mittal, and S. Rahimi, "Towards a HIPAA Compliant Agentic AI System in Healthcare," *arXiv preprint arXiv:2504.17669*, 2025.
9. Z. Huang et al., "ADAM-1: An AI Reasoning and Bioinformatics Model for Alzheimer's Disease Detection and Microbiome-Clinical Data Integration," *IEEE Access*, 2025.
10. X. Xu and R. Sankar, "Large Language Model Agents for Biomedicine: A Comprehensive Review of Methods, Evaluations, Challenges, and Future Directions," *Information*, vol. 16, no. 10, p. 894, 2025.
11. F. Neha, D. Bhati, and D. K. Shukla, "Retrieval-Augmented Generation (RAG) in Healthcare: A Systematic Literature Review," *AI*, vol. 6, no. 9, p. 226, 2025.
12. M. T. Zamir et al., "Explainable AI-Driven Analysis of Radiology Reports Using Text and Image Data: Experimental Study," *JMIR Formative Research*, vol. 9, no. 1, e77482, 2025.
13. P. S. Khade, "Explainable AI (XAI) in Healthcare: Building Trust in Medical Diagnosis Systems," *Int. J. Sci. Res. Sci. Technol.*, 2025.

14. H.-S. Lee et al., "The Ethics of Simplification: Balancing Patient Autonomy, Comprehension, and Accuracy in AI-Generated Radiology Reports," *Research Square*, 2025.
15. Z. He et al., "Quality of Answers of Generative Large Language Models vs Peer Patients for Interpreting Lab Test Results for Lay Patients: Evaluation Study," *JMIR*, 2024.
16. E. Steimetz et al., "Use of Artificial Intelligence Chatbots in Interpretation of Pathology Reports," *JAMA Network Open*, vol. 7, no. 5, e2411578, 2024.
17. I. Malashin et al., "Image Text Extraction and Natural Language Processing of Unstructured Data from Medical Reports," *Machine Learning and Knowledge Extraction*, vol. 6, no. 2, p. 64, 2024.
18. J. Ji et al., "Vision-Language Model for Generating Textual Descriptions From Clinical Images: Model Development and Validation Study," *JMIR Formative Research*, vol. 8, no. 1, e38330, 2024.
19. K. C. Anderson et al., "Natural Language Processing for Healthcare (BioBERT and ClinicalBERT evaluation)," *Various*, 2023.
20. A. S. Zamir et al., "Controlling AI Hallucinations in Clinical Workflows: A Systematic Review of RAG and Verification Techniques," *arXiv preprint arXiv:2311.05665*, 2023.
21. G. Damiani et al., "Potentiality of Algorithms and Artificial Intelligence Adoption to Improve Medication Management in Primary Care: A Systematic Review," *BMJ Open*, vol. 13, no. 3, e068845, 2023.
22. M. Panda and S. R. Mahanta, "Explainable Artificial Intelligence for Healthcare Applications Using Random Forest Classifier with LIME and SHAP," in *Explainable AI in Healthcare*, Springer, 2023.
23. E. J. Topol, "Role of Generative AI in Patient-Doctor Communication," *Substack*, 2023.
24. S. Han and W. Choi, "Development of a Large Language Model-based Multi-Agent Clinical Decision Support System for Korean Triage and Acuity Scale-Based Triage and Treatment Planning in Emergency Departments," *OAJAIML*, 2023.
25. D. Varshney et al., "Knowledge Grounded Medical Dialogue Generation Using Augmented Graphs," *Scientific Reports*, vol. 13, p. 29213, 2023.